

IMPLEMENTATION PLAN

Carbon Horizon

Learn Your Impact. Shape the Future.

Version 1.0

Project Goal

Build and deploy a production-ready MVP of Carbon Horizon that demonstrates:

- Carbon Footprint Assessment
- AI Sustainability Coach
- Carbon Reduction Simulator
- Carbon Horizon Forecast
- Habit Tracking
- Cloud Deployment

Target Platform:

Google Cloud

Repository:

GitHub

Development Methodology:

Agile + Incremental Delivery

Development Roadmap

Phase 0

Project Setup

↓

Phase 1

Authentication



Phase 2

Carbon Assessment



Phase 3

Dashboard



Phase 4

AI Sustainability Coach



Phase 5

Carbon Reduction Simulator



Phase 6

Carbon Horizon Forecast



Phase 7

Habit Tracker



Phase 8

Testing



Phase 9

Cloud Deployment



Phase 10

Competition Demo Preparation

Phase 0

Project Initialization

Duration:

1 Day

Objectives:

Create project foundation.

Tasks:

Create GitHub Repository

Setup FastAPI Project

Setup Frontend Structure

Configure Git Workflow

Create README

Create Project Board

Setup Environment Variables

Deliverables:

Repository Ready

Development Environment Ready

Recommended Git Structure

main

Production

develop

Development

feature/authentication

feature/carbon-calculator

feature/dashboard

feature/ai-coach

feature/simulator

feature/forecast

feature/habits

feature/deployment

Phase 1

Authentication Module

Duration:

2 Days

Objectives:

Enable user registration and login.

Tasks:

Create User Model

Password Hashing

JWT Authentication

Login API

Registration API

Protected Routes

Testing

Deliverables:

Working Authentication System

Phase 2

Carbon Footprint Assessment

Duration:

3 Days

Objectives:

Calculate user emissions.

Tasks:

Create Carbon Assessment Model

Create Emission Input Model

Create Carbon Factors Dataset

Implement Calculation Engine

Build Assessment Form

Generate Results

Deliverables:

Carbon Formula Module

Transportation

Distance × Emission Factor

Energy

kWh × Emission Factor

Food

Diet Factor × Frequency

Waste

Waste Score × Factor

Output:

kg CO₂e

Phase 3

Dashboard Development

Duration:

2 Days

Objectives:

Visualize carbon footprint.

Tasks:

Create Dashboard APIs

Build Chart.js Visualizations

Emission Breakdown

Carbon Score

Trend Graphs

Deliverables:

Interactive Dashboard

Phase 4

AI Sustainability Coach

Duration:

3 Days

Objectives:

Integrate Gemini.

Tasks:

Gemini API Setup

Prompt Engineering

Context Builder

Conversation Storage

Recommendation Engine

Deliverables:

AI Coach Functional

AI Prompt Structure

Inputs:

User Footprint

Lifestyle

Goals

Simulation Results

Output:

Reduction Recommendations

Educational Insights

Action Plan

Phase 5

Carbon Reduction Simulator

Duration:

3 Days

Objectives:

Enable scenario simulation.

Tasks:

Create Simulation Model

Simulation Engine

Comparison Logic

Reduction Calculations

Visualization

Deliverables:

Working Simulator

Example:

Current

3.2 Tons



Switch To Bus



2.4 Tons



25% Reduction

Phase 6

Carbon Horizon Forecast

Duration:

3 Days

Objectives:

Predict future emissions.

Tasks:

Forecast Engine

Scenario Builder

Trend Generation

Forecast Charts

Comparison Dashboard

Deliverables:

MVP Forecast Logic

Current Emissions

+

Habit Trend

+

Simulation Data

=

Future Projection

Phase 7

Habit Tracker

Duration:

2 Days

Objectives:

Track sustainable behavior.

Tasks:

Habit Model

Habit APIs

Streak Calculation

Progress Tracking

Dashboard Integration

Deliverables:

Habit Tracking Module

Phase 8

Testing

Duration:

2 Days

Objectives:

Validate application.

Tasks:

Unit Testing

API Testing

Authentication Testing

Calculation Validation

Simulator Testing

Forecast Testing

UI Testing

Deliverables:

Stable Application

Testing Coverage

Authentication

100%

Carbon Calculator

100%

Simulator

100%

Forecast

90%

Dashboard

90%

AI Coach

80%

Phase 9

Deployment

Duration:

2 Days

Objectives:

Deploy on Google Cloud.

Tasks:

Dockerize Application

Create Dockerfile

Create Cloud Run Service

Configure Secrets

Deploy Backend

Deploy Frontend

Configure Domain

Deliverables:

Live Application

Google Cloud Architecture

Frontend

Static Website

↓

Cloud Storage

or

Cloud Run

↓

FastAPI Backend

↓

SQLite/PostgreSQL

↓

Gemini API

Phase 10

Competition Preparation

Duration:

2 Days

Objectives:

Prepare final presentation.

Tasks:

Create Demo Script

Create Architecture Diagram

Create Workflow Diagram

Create Pitch Deck

Record Demo Video

Prepare Backup Dataset

Deliverables:

Competition-Ready Project

GitHub Project Management

Columns

Backlog

To Do

In Progress

Testing

Done

GitHub Issues

1 Setup Project

2 Authentication

3 Database Schema

4 Carbon Calculator

5 Dashboard

6 Gemini Integration

7 Simulator

8 Forecast Module

9 Habit Tracker

10 Deployment

11 Documentation

12 Presentation

Definition of Done

A feature is complete when:

Code Implemented

Tests Passing

API Documented

UI Integrated

Reviewed

Merged into Develop Branch

Final MVP Scope

Included

- ✓ Authentication
- ✓ Carbon Assessment
- ✓ Dashboard
- ✓ AI Coach
- ✓ Simulator
- ✓ Forecast
- ✓ Habit Tracker
- ✓ Deployment

Not Included

- ✗ OCR Bill Scanner
- ✗ Mobile App
- ✗ Community Features
- ✗ IoT Integration
- ✗ Leaderboards
- ✗ Carbon Marketplace

These features belong to Version 2.

Estimated Timeline

Phase 0 → 1 Day

Phase 1 → 2 Days

Phase 2 → 3 Days

Phase 3 → 2 Days

Phase 4 → 3 Days

Phase 5 → 3 Days

Phase 6 → 3 Days

Phase 7 → 2 Days

Phase 8 → 2 Days

Phase 9 → 2 Days

Phase 10 → 2 Days

Total:

Approximately 23 Working Days

(3–4 Weeks for a Student Team)

Success Criteria

The project will be considered successful if:

- Users can calculate their carbon footprint.
- AI generates personalized sustainability guidance.
- Simulations demonstrate measurable reduction opportunities.
- Forecasts visualize future environmental impact.
- Users track eco-friendly habits.
- The application is deployed successfully on Google Cloud.
- The complete workflow can be demonstrated in under 5 minutes during evaluation.